TOS

TOS

2025

– TOS : , , , . (AGI) . , AGI (2025–2500) (2025–2100), , , / –AI . , , , , TOS.[1]

[1] , “” TOS.

1

TOS — , . AGI .
(30-day rolling, updated hourly at metrics.tos.network):

- /: 847 (, 2025)
- : 12.4 (→)
- : 1.8%
- : 99.2% (7- -Sybil)

1.1

12–24 AI AGI Work (AGIW), TOS (TEM) . , – 2025 ., 2,847 devnet testnet, 126 97.8% .[2] AGIW 83% 99.2% ; -Sybil 112 . , 60–80% , . Q4 2026 . 5,000 , 10,000 , 15 2%.

100- Power of AI (PAI), / , AGI , , . Bloomberg Intelligence AI $1.3 2032 .;[3] IDC , $38 2024 . $121 2028 .[4] McKinsey , AI $2.6–$4.4 .[5] , , –AI . TOS , , AI 22- .

1.2

, AI :

- AI-Native ($121 2028 .): , AI , , , , . TOS , .
- ($850 2035 .): AI , (CC) (EC) , . TEM , .
- ($2.5 2045 .): AGI , , DID TOS, . TOS , .

TOS (0.1–0.5%), (12%), (8%) , , . $6.1 2030 ., 50,000 $300.

1.3

- : (DIDs), , , RepGraph.
- AGIW: , .
- / (CC/EC): , TEM .
- Power of AI (PAI): AI , 10,000+ TPS .
- : , , , .

1.4

1. : , , .
2. , .
3. : , , , .

[2] TOS, "AGI Work QA Report AGIW-2025-09," September 2025.
[3] Bloomberg Intelligence, "Generative AI Market Outlook," June 2024.
[4] IDC, "Worldwide Autonomous Agent Platforms Forecast," April 2024.
[5] McKinsey Global Institute, "The economic potential of generative AI," June 2023.

2 :

2.1

“TopoSpartan” : , TOS . “Topo” BlockDAG. , ( ), TOS (DAG), . , . , , 2,500+ (TPS) , 10,000+ TPS Power of AI (PAI).[6] DAG : , , .

“Spartan” , - . , —, . AGIW ; ; , . , , TOS : , , . : TEM (TOS) , , Sybil, . , “Topo” “Spartan” , , — , .

2.2 ,

Economist , "AI , , , .[7] Forbes , , AI " , .[8] Gartner , 2026 . 30% ,[9] Accenture , 40% CTO .[10] McKinsey 1,684 , , — , TOS .[11]

: , , AI , . , . , AI . 2 , AI , . (.), , . TOS , , .

2.3

AGI , , . , , . .

MERCURY	AI , ,
VENUS	,
MARS	,
JUPITER	, ,
SATURN	Dyson-swarm,
URANUS	,
NEPTUNE	, ,
PLUTO	,
PROXIMA	,
ANDROMEDA	,

, TOS . MERCURY , . VENUS , . MARS JUPITER , –AI . , SATURN ANDROMEDA , , .

: AGI. , TOS , . :

AGI	(")	(")

, DID, , , ,

[6] TOS, “BlockDAG Performance Benchmark PERF-2025-08,” August 2025.
[7] The Economist, “Meet the agentic future,” July 13, 2024.
[8] Forbes, “Why Enterprises Need Verifiable AI Agents,” August 19, 2024.
[9] Gartner, “Top Strategic Technology Trends 2025,” October 2024.
[10] Accenture, “Autonomous Agents Are Reshaping Enterprise Automation,” August 2024.
[11] McKinsey Global Institute, “The economic potential of generative AI,” June 2023.

AGIW , ,	, ,
CC/EC , , TEM	Dyson-swarm, ,
, ,	, , –AI
PQC, ,	, ,

(, , ,), , , , , , , , . : .

3

3.1

, :

1: (2015–2020) (AlphaGo DeepMind, Dota 2 OpenAI) (GPT-2, BERT). , . : .

2: (2020–2023) GPT-3 (2020 .) ChatGPT (2022 .) AI. : AutoGPT, LangChain Semantic Kernel LLM . Microsoft Google Copilot" . AlphaCode DeepMind . : , .

3: (2023–) AI (), (SQL), ( ) ( ). IDC , $38 2024 . $121 2028 .[12] Accenture , 40% CTO , - .[13] McKinsey 1,684 , 46% , , — , TOS .[14] Economist , , , AI, , .[15] Forbes , AI " .[16] : — , , AI .

3.2

AI . , :

, , . .

, AI . : (1) : 10–100× , , ; (2) : ; (3) AI : , ; (4) : , . 2 , AI , .

, AI pioneeri AI- . AI , : (1) , ; (2) , ; (3) , .

, . ; . ( ), ( ).

GPU/CPU . , : (1) (DIDs,); (2) ; (3) .

[12] IDC, "Worldwide Autonomous Agent Platforms Forecast," April 2024.
[13] Accenture, "Autonomous Agents Are Reshaping Enterprise Automation," August 2024.
[14] McKinsey Global Institute, "The economic potential of generative AI," June 2023.
[15] The Economist, "Meet the agentic future," July 13, 2024.
[16] Forbes, "Why Enterprises Need Verifiable AI Agents," August 19, 2024.

3.3

Ball et al. (. zkSNARKs) .[17] (1) , (2) , (3) ". TOS , (BlockDAG) (AGIW), .

Intel SGX, AMD SEV-SNP ARM TrustZone : , . TEEs AGIW, . ; TOS zk-SNARK .

(W3C DIDs, VCs) W3C . TOS , (Active Directory, OAuth) (eIDAS, GDPR).

3.4 TOS

TOS :

- TopoSpartan (BlockDAG +) .
- AGIW (AGI Work) TEE .
- CC/EC , , .
- TEM/PAI (TOS / Power of AI) AI .
- DID , , Web2 Web3 .
- , .

. TOS , .

4

4.1

TOS . , .[18]

1. 0: – BlockDAG Power of AI (PAI) .
2. 1: – (DIDs), (VCs) (RepGraph).
3. 2: – AGI Work (AGIW), / (CC/EC), TOS (TEM).
4. 3: – , , , .

- : , , .
- : API , , AML/KYC.
- : 24/7 SOC ( ), , .
- : SDK (Python, JavaScript, Rust), CLI , testnet, .

4.2 0:

BlockDAG TOS (DAG), 1–3 . : . DAG (2,500+ TPS , 10,000+ TPS PAI).[19] ; , .

[17] Ball et al., "Proofs of Useful Work," IACR ePrint 2017/203.
[18] explorer.tos.network.
[19] TOS, "BlockDAG Performance Benchmark PERF-2025-08," August 2025.

libp2p (, NAT,). . Mempool , , . (, ,) gRPC .

, Merkleised state trie (Patricia Merkle Trie). , .

4.3 1:

DID (DID), W3C. DID , (. , URI). registerDID(didDocument), . DID. ; .

(VCs) (, ,) VCs, DID. VCs (IPFS, Arweave) , . : 1000 ". , .

Merkleised : Merkle ID . (Merkle) , . .

RepGraph ( ) $r_a \in [0, 1]$, , , . ??. , . Sybil , (v4.0) ( ).

, Pedersen Bulletproof. , (, ,). , .

4.4 2: AGIW

Task Factory TaskInstance. (, , , ,). , (. 0.01 TOS) . , .

Task Instance . : Open → Claimed → Submitted → Validating → Settled (Disputed). claimTask(), ( ), submitWork(resultHash, attestation), . , .

Validation Pool TOS, . 3–7 ( ). (. , ,) validateWork(taskID, score) . ; (>2) . : $Acc_{t+1} = 0.9 \times Acc_t + 0.1 \times correctness_t$.

Reward Splitter $q \in [0, 1]$: $e \times q$, $e \times (1 - q) \times \beta$, $e \times \gamma$. : $\beta = 0.12$, $\gamma = 0.08$. . (5–60), .

. , . ( ) , . ; . (TEM) .

4.5 3:

CC/EC (CC) TFLOP-; (EC) kWh. , AWS, Azure, GCP (CC) ISO (EC). CC/EC TOS , , .

21 ( ) TOS 15 . , . (30) . (±5%) , .

TEM TOS , , . , 2%, TEM , . CC/EC (. 20%), TEM (AMM). TEM .

(AMM) CC/EC ↔ TOS. (0.3%); TEM . (Uniswap v3), .

4.6 4:

(DSL) . : “ >10,000 TOS .” Python (, ,). , .

(. “ ,” “ 3/5 ”). (75%) . “”, , .

(7–15) . ( ), (,) . , .

(, ,) . , . (. AGIW,) API , .

4.7

(IDS) gRPC, Apache Arrow (,) GraphQL API. , (. ,) ; , . IDS TOS Explorer, .

API AGIW, . : “ ,” “ , ,” “ CC/EC .” , OAuth2, .

SDK Rust, Python, TypeScript . : tos-sdk-py Agent.register_did(), Task.publish(), Task.claim(), Task.submit(). Terraform . GitHub Actions CI/CD .

4.8 ()

:

1. ( )
   . , DID , DIDRegistry.register().
   . KYC , KYC-Verified.
   . TOS (. 100 TOS), ; 7 .
2. ( )
   . (. ,).
   . ; .
3. ( )
   . TaskFactory , .
   . TaskInstance.claim() (≥ ,).
   . Claimed; ; .
4. (+)
   . (TEE): Intel SGX, AMD SEV-SNP ARM Realm.
   . TEE : ( ), /, , .
   . , , submitWork().
5. ( )
   . ValidationPool 3–7 (,).
   . , , .
   . (), , , .

6. (+)

   . RewardSplitter : $e \times q$ CC/EC, , .

   . RepGraph : $r_{t+1} = 0.9 \times r_t + 0.1 \times q$.

   . , ; ArbitrationContract.

7. ( )

   . (+).

   . (. PQC).

   . , .

4.9 :

1. ( ) : “ PII 500- .” : 50 CC () + 5 EC (). : 2 .
2. A (DID did:tos:tos1qpz8vq6qgnkgw9zzq9z5qgpgqx8qgz5dpgrxve, 0.87, Certified Legal AI) . : Claimed.
3. A Intel SGX . , (4.8 EC), (22).
4. A ( ) + . : Submitted.
5. (3): , 10% , . : 0.95, 0.94, 0.96. : 0.95.
6. RewardSplitter A $50 \times 0.95 = 47.5$ CC + $5 \times 0.95 = 4.75$ EC. 6 CC + 0.6 EC. 4 CC + 0.4 EC.
7. RepGraph A: $r_{t+1} = 0.9 \times 0.87 + 0.1 \times 0.95 = 0.878$.
8. , , . (30 GDPR).

   , , , AI .

5 ( ) –

5.1

, . : 30% , $120 .[20] , . , , , .

5.2 12–18

D1. v1

- DID, BLS , DID , on-chain .
- , (VCs), , ; , Merkleised .
- Pedersen Bulletproof; , .

D2. AGI (AGIW) v1

[20] Gartner, “Top Strategic Technology Trends 2025,” October 2024; IDC, “Worldwide Autonomous Agent Platforms Forecast,” April 2024.

- Task Factory Task Instance , , .
- Intel SGX, AMD SEV-SNP ARM Realm; on-chain.
- Validation Pool ; slashing.
- Reward Splitter CC/EC , .

D3. /

- CC, TFLOPs; EC, kWh. .
- Paymaster CC/EC; TEM (.).

D4.

- , , , AGIW, .
- Explorer , , , , .
- SDK Rust, Python, TypeScript; Terraform .

5.3

	(12)	
	5.000	DID , , paymaster
	10.000	AGIW , ,
	< 15	on-chain,
	< 2%	
CC/EC	> 60%	TEM (,)
	60–80% - SOC 2	API ,

5.4 ( )

DID DID Python (DIDRegistry.py) :

```
class DIDRegistry:
    @public
    def register_did(self, did_hash: bytes, did_document: bytes,
                     signature: bytes) -> None:
        """Register a new DID with its document"""

    @public
    def update_did(self, did_hash: bytes, new_document: bytes,
                   signature: bytes) -> None:
        """Update an existing DID document"""

    @public
    def revoke_did(self, did_hash: bytes, signature: bytes) -> None:
        """Revoke a DID"""

    @public
    @view
```

```
def resolve_did(self, did_hash: bytes) -> bytes:
    """Resolve a DID to its document"""
```

DID W3C DID Core (v1.0). :

- id: (. did:tos:tos1qvqsyqcyq5rqwzqfpg9scrgwpugpzysnzs23v9xhyepcg...)
- verificationMethod: , (assertion)
- service: , , API
- created, updated:

: 80,000 ($0.50 50 gwei, $2,000 TOS), .

Verifiable Credentials Credential issuance follows the W3C VC Data Model (v1.1). Issuers sign JSON-LD documents containing:

```
{
  "@context": ["https://www.w3.org/2018/credentials/v1"],
  "id": "https://issuer.example/credentials/987",
  "type": ["VerifiableCredential", "SkillCertificate"],
  "issuer": "did:tos:tos1qvqsyqcyq5rqwzqfpg9scrgwpugpzysnzs23v9xh",
  "issuanceDate": "2025-10-01T00:00:00Z",
  "credentialSubject": {
    "id": "did:tos:tos1qq5hqy2jwn0r5ehvuew5r3qjcfvjvq5dhldxvyq",
    "skill": "Data Science",
    "proficiencyLevel": "Expert"
  },
  "proof": {
    "type": "BbsBlsSignature2020",
    "created": "2025-10-01T00:00:00Z",
    "proofPurpose": "assertionMethod",
    "verificationMethod": "did:tos:tos1qvqsyqcyq5rqwzqfpg9scrgwpugpzysnzs23v9xh#key-1",
    "proofValue": "..."
  }
}
```

: VC (IPFS, Arweave); (32). VC, .

Revocation Registry Merkle 2^{256} . ; ID . Merkle , , ID ( ). , . : 50,000 ($0.30).

Confidential Transactions Pedersen :

$$C(v, r) = vG + rH \tag{1}$$

v (), r , G H . : $C(v_1, r_1) + C(v_2, r_2) = C(v_1 + v_2, r_1 + r_2)$, , . Bulletproof $v \in [0, 2^{64})$, . : 672 , . : 1.2 ms . ; enableConfidentialMode() .

5.5 AGIW ( )

AGIW :

1. TaskFactory (TaskFactory.py)

: .

:

- publishTask(metadata, escrow, deadline, allowedAgents): TaskInstance, , 0.01 TOS, TaskPublished. ID .
- listTasks(filters): ; , , .
- updateWhitelist(agentDIDs): (. KYC-, > 0.7).

:

- : 0.01 TOS (,).
- : .
- : 150,000 ($0.90 50 gwei).

: ; RewardSplitter .

2. TaskInstance (TaskInstance.py)

: .

:

$$\text{Open} \xrightarrow{\text{claim()}} \text{Claimed} \xrightarrow{\text{submitWork()}} \text{Submitted} \xrightarrow{\text{validate()}} \text{Validating} \xrightarrow{\text{consensus}} \text{Settled} \tag{2}$$

: $\xrightarrow{\text{dispute()}}$ Disputed $\xrightarrow{\text{arbitration}}$ Resolved.

:

```
@dataclass
class Task:
    task_id: bytes              #
    publisher: str              #
    claimed_by: str             #   ,
    escrow: int                 #
    deadline: int               #
    spec_hash: bytes            # IPFS CID
    result_hash: bytes          #
    attestation: bytes          #     TEE
    state: TaskState            # enum: Open, Claimed, Submitted, ...
    validator_scores: dict[str, int]  #  [0,100]
    final_quality: int          #
```

:

- claimTask(agentDID): (1) Open, (2) , (3) $\geq$. Claimed, .
- submitWork(resultHash, attestation): (1) , (2) , (3) . Submitted, WorkSubmitted, .
- validateWork(score): . $\in [0, 100]$. , , Settled, RewardSplitter.

:

- , forfeit(), 10% ( ). 0.05.
- 24 , : ; 5% .

3. ValidationPool (ValidationPool.py)

: , , .

:

- 1,000 TOS, .
- Acc_v 0.5, :

$$Acc_{v,t+1} = \alpha \times Acc_{v,t} + (1-\alpha) \times correctness_t, \quad \alpha = 0.9 \tag{3}$$

$correctness_t \in \{0,1\}$ ($\pm 10\%$).

:

1. WorkSubmitted, ValidationPool $n = 5$ ().
2. :

$$P(\text{select } v) \propto stake_v \times (1 + Acc_v) \tag{4}$$

3. ; 24 .

:

- : $> 2\sigma$, 1% .
- : 5% .
- : > 10 ( ), ; 20% .

: $e \times \beta \times (1-q)$, e , $\beta = 0.12$, q . Acc_v.

4. RewardSplitter (RewardSplitter.py)

: .

: e (CC/EC) $q \in [0,1]$:

$$R_{\text{agent}} = e \times q \times (1 + \alpha(r_a - 0.5)) \times f(s_a) \tag{5}$$

$$R_{\text{validators}} = e \times \beta \times (1-q) \tag{6}$$

$$R_{\text{treasury}} = e \times \gamma + \tag{7}$$

:

- $r_a \in [0,1]$ (??),
- $\alpha = 0.2$,
- $f(s_a) = \min(2, 1 + \log(1 + s_a))$ ( ),
- $\beta = 0.12$,
- $\gamma = 0.08$.

: (. ??):

- $(r \geq 0.90)$: 5 .
- $(0.70 \leq r < 0.90)$: 15 .
- $(0.30 \leq r < 0.70)$: 30 .
- $(r < 0.30)$: 60 .

Sybil , .
: RewardSplitter CC/EC, ~30,000 ($0.18).

5.6

CC/EC , 21 . TOS , API (, ISO). . , . , .

5.7

:

- gRPC , Apache Arrow.
- , .
- (Grafana), , , .
- API , .

5.8

T , A V . t c_t e_t. U_a :

$$U_a = \sum_{t \in T_a} q_{a,t} \cdot e_t - \gamma s_a - p_a \tag{8}$$

$q_{a,t}$, s_a , p_a γ . U_v . :

$$\sum_{a \in A} s_a = \theta \sum_{t \in T} e_t, \quad 0.2 \leq \theta \leq 0.4 \tag{9}$$

- .

:

$$Reward_a(t) = e_t \times q_{a,t} \times (1 + \alpha(r_a - 0.5)) \times f(s_a) \tag{10}$$

$f(s_a) = \min(2, 1 + \log(1 + s_a))$ r_a . $Reward_v(t) = e_t \times \beta \times Acc_v(t)$, Acc_v . δs .

5.9 (RepGraph)

RepGraph , (, ,) (,).

r_a^{base} a :

$$r_a^{\text{base}} = 0.3 \times \frac{\text{age}_a}{\text{age}_{\max}} + 0.4 \times \frac{\text{txCount}_a}{\text{txCount}_{\max}} + 0.3 \times \frac{\text{stake}_a}{\text{stake}_{\max}} \tag{11}$$

:

- age_a (365),
- txCount_a ,
- stake_a TOS,
- .

q_t, Acc_a :

$$Acc_{a,t+1} = 0.9 \times Acc_{a,t} + 0.1 \times q_t \tag{12}$$

± 0.2:

$$r_a^{\text{accuracy}} = r_a^{\text{base}} + 0.2 \times (2Acc_a - 1) \tag{13}$$

$(Acc_a \to 1 \Rightarrow +0.2)$ $(Acc_a \to 0 \Rightarrow -0.2)$.

, $Acc_a > 0.85$ > 100 , $+0.1$, $r_a = 1.0$.

r_a, , :

$r \geq 0.90$	5	30–50%	14%
$0.70 \leq r < 0.90$	15	10–30%	33%
$0.30 \leq r < 0.70$	30	0–10%	41%
$r < 0.30$	60	: 2×	12%

4: RepGraph 2025.

Sybil RepGraph Sybil :

1. : 100 TOS, 7 ( ).
2. : (< 7) (< 10 CC).
3. IP: , IP ; .
4. : , , .
5. : (KYC,).
6. (v4.0): BrightID, Worldcoin Gitcoin Passport, - .
7. : RepGraph ; .

2025 . 112 0.7% .[21]

5.10

(SLO) :

- SLO1 – : 99.9% AGIW 20 (p99.9 < 20).
- SLO2 – : CC/EC 15 ; (> 30) .
- SLO3 – : < 2%; , TEM 20%, .
- SLO4 – : ≥ 99.95% (< 4.4 /).
- SLO5 – : ( ).

:

- : (//, ,) gRPC (IDS).
- : IDS Apache Arrow (10× ,).
- : (, ,) SHA-256 ; .
- : Grafana , , , , .
- API : RESTful (OAuth2-,) : “ AGIW ,” “, ,” “ CC/EC .”

Prometheus SLO, (. TEM,).

[20] TOS, “Reputation & Anti-Sybil Telemetry REP-2025-08,” 2025.
[21] TOS, “Reputation & Anti-Sybil Telemetry REP-2025-08,” 2025.

5.11

10,000 ($\in [0,1]$, $\in [100, 10000]$ TOS) :

- : ± 0.05 30 (95%).
- : ($r < 0.3$) ($r > 0.7$) 50 , .
- : , 42% ; (5–60), 6%.
- : 10% 28% (0.34), ; α ( ) $f(s_a)$ ( ), .

5.12

AGIW devnet (Aurora) testnet (Helios) – 2025 . .

(15 – 15 2025 .)

- : 2,847 (, ,).
- : 126 (, ,).
- : 97.8% (123/126 ; 3 ,).
- : 83% (: 12.3).
- : 99.2% (112 , 1).
- : 60–80% .

(1,000 , 2025 .)

- : 7.4 ( ).
- : 3.1% (TEE + 230 ms).
- : 0.002 TOS (+).
- : 45 / (2,700 /) .
- : 50 (,); , 100% .

5.13

TOS Foundations AI:

- US AI RMF (NIST): , , , (, ,).
- EU AI Act: AI (,) TOS KYC, , .
- MAS : , AI , ; TOS API , MAS.

AI , TOS :

- : , AI (12%).
- : , ( ).
- : / ( ).
- : ( ).

(??) , 10 TOS 3–4 , .

EC , . "; , / , 10% (TEM). , .

6 –

6.1

, AI , .[22] , , .

6.2

, :

1. (CustodyVault.py)
: , . .
:

- - : m--n , m (, ,) , T. : $> 10,000$ TOS 2--3 (+ , + , +).
- (HSM): FIPS 140-2 Level 3 HSM (AWS CloudHSM, Thales Luna). HSM ; .
- : , :
  - : $X //.
  - : (,).
  - : (OFAC).
  - : .
- : , .

: SOC 2 Type II HSM, , . (Active Directory, Okta) SAML/OAuth.

2. (BridgeController.py, FiatGateway.py)
: (, ,) .
:

- HTLC: - TOS . : 100 TOS TOS H; H; , , 24 .
- : - (7--11) ; ($< \$1M$) . (TLA+) , .

:

- API: ISO20022 (SWIFT, FedNow, SEPA). USD/EUR; CC/EC .
- : KYC/AML Chainalysis, Elliptic; , (>95%), .
- : 2–4 ( ); 24–48 ( ).

:

- : CC (GPU). : 20% , 10% .
- : EC ; / .
- : , ; (3%) .

3. (ArbitrationCourt.py)
: , (,), .
:

- : (, AI,), TOS 10,000 TOS.

[22] McKinsey Global Institute, "The economic potential of generative AI," 2023.

- : (, , ,); .
- : ; (< 3.5/5 20) .

:

1. : ( ) (5–10%).
2. : 3 (, ,).
3. : (, ,) IPFS; .
4. : , , .
5. : (2--3) (, ,). , .
6. : ; (+).

: TEM (2% AGIW); , , ( ). 4--7 .

4. (SafetyOracle.py)
: AI (, ,); .
:

- AI: AI Anthropic, GPT OpenAI, Google DeepMind.[23]
- API : Perspective API (), AWS Comprehend (,), Microsoft Azure Content Safety.
- : , , , (6 GDPR).

:

- : (> 0.9,), ; .
- : , (>3 30), (. 10 /).
- : , , GDPR ; .

: : , , (< 2%).

6.3

- , x. . , ; .

6.4

HTLC , . ISO20022 . USD ; , CC . CC/EC , .

6.5

?? .

,	( )
,	()
HSM , ,	( )
,	( )
,	( )

5: – .

[23] IBM Research, “AI Safety Metrics for Enterprise Deployment,” 2024 .

6.6 Power of AI (PAI)

PAI TOS BlockDAG (2,500+ TPS) AI , 10,000+ TPS, .

1: AI ( ) : AI ( ) . , (, AGIW,). $> 67\%$, AI; BlockDAG.
: ; . - : AI $\Rightarrow$ BlockDAG.
: 20–30% - .
: AI (Apache 2.0), , (,).

2: ( ) : :

- AI (30%): , AI, , AGIW.
- (70%): AI, (,), .

: k " $1 - 2^{-k}$. AI ; .
: TLA+ ; ( ) ( ) ($< 33\%$).
: 3–4× (7,500–10,000 TPS) .
: , AI (,). 4--7 .

3: PAI (/) : AI :

- : , (DDoS, Sybil) .
- : -DAG () ; AI .
- : AI , TEM , .

: 10,000+ TPS , 15,000+ TPS , 95% .
: :

- : 500 ( ).
- : < 0.45 ( ).
- AI : , , .

: PAI (??), (, /) (- , AGI).

6.7

( ) AGIW, (HSM, -), / , AI. (60–80%) (SOC 2, EU AI Act) .

( )
(DAE), , . PAI , –AI , .

6.8

	, (1,000 TOS), EWMA	(11/21 ;)
	, , , - RepGraph	(;)
( )	HSM (FIPS 140-2 Level 3), - ,	(HSM +)
	, ,	
	( ), (ML),	(; TEM)

( )	(Anthropic, OpenAI, Google, (;)), ,
PAI	AI , (. , .), , (;) (TLA+), - (7--11), (4+ +) ($1M/tx), (3%)

6: – (-).

7 –

7.1 -

2070- 1.0 . AI . , .

7.2

, . , -. .

1. Policy DSL (,): . :

```
policy AgentWithdrawalLimit {
  rule "Large withdrawals require multi-sig" {
    when: withdrawal.amount > 10000 TOS
    require: signatures.count >= 2
    from: [agent.key, guardian.key, auditor.key]
    timeout: 24 hours
  }
  rule "Sanctioned addresses blocked" {
    when: transfer.recipient in SanctionsList
    action: reject
    log: "OFAC violation attempt"
  }
}
```

2. :
   - : DSL, (AST).
   - : ( ), ( ), .
   - : Python , TOS.
   - : , .
   - : SMT (Z3, CVC5) (. “ ,” “ ”).
3. :
   - DSL, GitHub.
   - CI/CD : , testnet, .
   - , , (7-).
   - : 60% , 75% .
   - : 48- ( ).

:

- : , ( ).
- : , , (onlyAgent, onlyGuardian, onlyGovernance).
- : , .
- : ; ; .

- : “ X , Y” (ITAR).
- : “ 1,000 EC/ ; .”
- : “ > 10 1 , .”

7.3 Delay-Tolerant Settlement Protocol

, CRDT, . ?? .

Algorithm 1 Delay-Tolerant Settlement (DTS)

```
Input: Operation log L_i for participant i.
Broadcast digest d_i = H(L_i) periodically.
On receiving digest d_j, request missing operations.
Merge logs via L_i ← merge(L_i, L_j), ensuring commutativity.
When quorum acknowledges, emit finality proof to main chain.
```

7.4 Metrics

Governance SLOs: 75% participation on critical votes, proposal lead time $\geq$ 48 hours, zero emergency rollbacks per quarter. Telemetry includes vote turnout, execution success, dispute outcomes.

7.5 Use Cases

- , .
- -AI , .
- .

8 –

AGI ( ) ( ), TOS , , . , TOS - .

8.1

: - 8 44 . 2.5 . (,) 40–80 . (BFT, PoS) ; :

- : .
- : (,).
- : .

: TOS :

1: (LCD)

(, , , .) TOS . LCD BlockDAG + PAI ( ). ; .

2: (IRC)

24 ( ), LCD IRCMerkle zk-SNARK , . IRC :

1. BFT : Tendermint ; 60 .
2. : 51% (67%), - .
3. : 7 ( ), IRC , LCD.

. :

1. HTLC (,) LCD, H 48- .
2. LCD HTLC IRC ( ).
3. LCD IRC, HTLC, - x ($H = \text{hash}(x)$).
4. LCD IRC.
5. LCD , ( ).

: = 48–96 (15). - .

: 40% , LCD IRC.

: IRC Merkle , , . LCD : , . : 50% ; LCD , .

( )

- TPS: 10,000+ LCD (PAI).
- : 24–48 ( ).
- : 7 ( ).
- : 30- (,).

8.2

: AGI (, ,), . :

- , ($50B+).
- .
- ( ) .

. TOS , .

:

24 , LCD :

- : Merkle , , .
- : zk-SNARK , (, ,).
- : , , , .

IRC ( ).

(, 75%):

1. : (, ,) .
2. : , (,), (RAC), :
   - ID (,).
   - (,).
   - (, .).
3. : RAC ; 75% + .
4. : , . ; .

- : ; (“ ”). .
- : 90- (-). .
- : 5% , (. ,).
- : 1 ; .

: (2185) : 100- ECDSA , $12B 5,000 8 (PQC).
:

1. (10,000).
2. ; 2 .
3. ; .
4. (82%) 12 .
5. ; ; .
6. $200M , .
7. : PQC ; Dilithium-5 30 ; .

: 7 ; ; .

8.3

: AGI , . :

- , , 200 .
- ( ).
- (/AI).
- (X Y 2234?).

(AWS Glacier, Google Coldline) <100- ; , , .

: 1: (1–10)
+ SSD/NVMe. . : 3 LCD, (: 3 ; : , ,).
2: (10–100)
HDD . Reed-Solomon (20%) 20% . ; . : 0.5 TOS TB- .
3: (100–1,000)
:

- DNA : (Church Lab, Harvard). : 215 ; : 2,000+ , -18ºC. : $1,000 TB (2025; $10/TB 2050).
- 5D : (University of Southampton). : 13.8 ( ). : 360 TB .

- (LTO-12+): . : 50–100 ( ).

: (7) ; (: , ; : ; :).
4:
:

1. ; ; .. ( ).
2. : Merkle , .
3. (.), , .

(. “ tos1qq5hqy2jwn0r5ehvuew5r3qjcfvjvq5dhldxvyq 2234-05-15?”):

1. (10): O(log n) , 100ms .
2. (10–100): , Merkle , 10–60 .
3. (100–1,000): / , / , Reed-Solomon, 1–7 .

: 3% . $50B (2150), $1.5B/ :

- ($500M/, 500 PB).
- 5D ($300M/, 1,500 PB).
- / ($400M/).
- ($300M/, 1,000).

8.4

- 2030-: 50- (ECDSA).
- 2040-: 1,000- , (RSA-2048, ECDSA-256).
- 2050-: , .

: 1 (v4.0, 2026): . :

- : ECDSA secp256k1 ( ).
- -: Dilithium-3 (NIST FIPS 204).

; , (, PQC).
2 (v5.0, 2030): PQC . ; ECDSA. : 12 .
3 (v6.0, 2040): . SHA-256 ( ) SHA-3 BLAKE3 (256-).

( ) , TOS PQC:

1. 1 (1–3): (, ,). 5,000+ PQC .
2. 2 (4): (MPC) . ; .
3. 3 (5–6): . PQC ; .
4. 4 (7): (67%). , mainnet 30 .
5. 5 (8): mainnet. 12 .

PQC (.):

- PQC.
- - (Falcon, SPHINCS+).
- .

8.5 :

: . ; TOS , . , TOS AGI :

- (,).
- (, ,).
- (,).

(90- , 75% ,), + . AGI , , .

9

TOS , . , .

9.1 P1 – Digital Personhood & Reputation

Decentralized Identifiers (DIDs) , W3C DID ( ), did:tos:.
DID Format:

did:tos:<bech32-address>
did:tos:<network-id>:<bech32-address>

Examples:
Simple form: did:tos:tos1qvqsyqcyq5rqwzqfpg9scrgwpugpzysnzs23v9xh
Mainnet: did:tos:mainnet:tos1qvqsyqcyq5rqwzqfpg9scrgwpugpzysnzs23v9xh
Testnet: did:tos:testnet:tst1qqqsyqcyq5rqwzqfpg9scrgwpugpzysnzs2pgde5k

: TOS bech32 DID , , . tos1, tst1.
DID :

- : Ed25519 (), X25519 (), Dilithium-3 (-).
- : HTTPS URL , IPFS .
- : (, ,).
- : Bloom , .

:

1. ; DID .
2. TOS (: 1 TOS).
3. DID ; DID IPFS; .
4. , .

Verifiable Credentials (VCs) , (, ,).
:

- : RepGraph (///), , .
- : (, ,), .
- : KYC/AML ( ), (EU AI Act, FDA).
- : (, DAO).

:

1. (. RepGraph) JSON-LD .
2. BBS+ (;).
3. DID IPFS; .
4. ; (. “ ≥ 0.7 ”).

Reputation Graph (RepGraph) Algorithms Score Computation:

$$r_t = \alpha \cdot r_{t-1} + (1 - \alpha) \cdot s_t \tag{14}$$

:

- r_t: t ([0, 1]).
- $\alpha = 0.85$: (85%).
- s_t: - (5).

Decay Mechanism:

$$r_{t+\Delta t} = r_t \cdot (0.98)^{\Delta t/30} \tag{15}$$

2% ; .
:

- : .
- : , , , .
- : (30) 2× .

:

- Sybil: 100 TOS + (BrightID v5.0).
- : ; .
- : 0.5 90 ; .

W3C

- DID Core v1.0: ; TOS DID Universal Resolver.
- v1.1: (TOS -TOS).
- DIDComm v2: - (, ,).

9.2 P2 – Economic Execution

Account Abstraction TOS , , . , AI .
Key Features:

- : (.).
- : m--n ( ).
- : ( ).
- : (,), .

UserOperation Structure:

```
{
  sender: "0x...",            // Agent's account address
  nonce: 42,                  // Sequential counter
  initCode: "0x...",          // Account creation code (if new)
  callData: "0x...",          // Function call to execute
  paymasterAndData: "0x...", // Paymaster address + verification data
  signature: "0x..."          // Signature over UserOperation
}
```

AGIW Settlement Engine , . ??.
:

- TaskInstance: , (→ → →).
- : 5 ; .
- : , TEE (Intel SGX, AMD SEV), .
- : , .

Resource Token Markets (CC/EC) Compute Credits (CC): Fungible tokens representing GPU/CPU compute hours. Pricing oracles aggregate AWS/GCP/Azure spot prices.

Energy Credits (EC): (kWh). ; 10% .
(AMM):

- (CPMM): $x \cdot y = k$ $x =$ CC, $y =$ TOS.
- 0.3% .
- : TEM , >1M TOS .

TOS Energy Model (TEM) , .
Fee Structure:

$$\text{Total Fee} = \text{Base Fee} + \text{Priority Tip} - \text{TEM Subsidy} \tag{16}$$

:

- , .
- : 50% ; 12.5% , > .

TEM Subsidy Categories:

- (100%): , , .
- (75%): , .
- (50%): / <$10M .
- (0%): .

: 8% TEM; .

7: TEM Economic Parameters and Control Mechanisms

Parameter	Value / Range	Control Logic
Base fee band (TOS)	0.0001–0.01 per gas unit	Adjust by block fill p95; target 50% utilization
Base fee adjustment	+12.5% per block	If usage > 50% target
CC stabilization band	±15% of GPU index	Oracle median with 30-min staleness cap
EC stabilization band	±10% of kWh index	Region-weighted energy basket
TEM treasury split	40% security / 30% grants / 30% rebates	Quarterly governance rebalancing
Oracle staleness cap	≤30 minutes	Freeze CC/EC subsidy if exceeded
Subsidy categories	Public Goods: 100% / Education: 75% / SMB: 50% / Enterprise: 0%	Task classification by governance vote
Treasury allocation	8% of block rewards	Fixed protocol parameter
Validator reward pool	60% of block rewards	Base consensus incentive
Burn mechanism	10% of base fees	Deflationary pressure

Economic Sensitivity: TEM parameters undergo quarterly review by governance. Simulation results (available at metrics.tos.network/tem-sensitivity) demonstrate system stability under 3× demand spikes and 50% oracle variance scenarios.

9.3 P3 – Consensus & Safety

BlockDAG Baseline Architecture ; TOS (DAG), .

:

- 1–3 (1).
- ; .
- ; + .

:

- : 8 (90).
- - : DAG, - .
- : 100 , (24); + .

:

- : 2,500–3,500 TPS (2025).
- PAI: 10,000+ TPS (Q4 2026).

8: Performance Evaluation Methodology

Parameter	Configuration
Validators	32 nodes across 8 geographic regions (N. America, Europe, Asia-Pacific, S. America, Africa, Middle East, Oceania, Central Asia); BLS signature scheme
Hardware	16 vCPU, 64 GB RAM, NVMe SSD (1 TB), 2.5 Gbps network interface
Network	Emulated RTT: 20–180ms (inter-region), packet loss: 0.1–1.0%, jitter: ±10ms
Workloads	Real agent jobs: code review (15%), data synthesis (25%), RAG tasks (20%); Synthetic spam: 20–60% injection rate
Runtime	72-hour continuous runs; 5 independent seeds; cold start + steady state phases
Block Size	Variable: 100–2,000 transactions per block; target 50% utilization
Mempool Config	Priority queue with reputation weighting; max 100,000 pending transactions
Metrics Collected	TPS (p50/p95/p99), block time (p50/p95), finality depth (confirmation blocks), AGIW settlement latency (task publish → reward), dispute rate (% of tasks disputed), validator liveness (uptime %), fork rate
Measurement Tools	Prometheus + Grafana; custom telemetry pipeline; raw logs exported to metrics.tos.network
Baseline Period	September 2025 testnet deployment (14-day observation window)

(2025):

- TPS: p50 = 2,847, p95 = 3,421, p99 = 3,789
- : p50 = 11.3s, p95 = 14.2s
- : 8 (90s)
- AGIW: p50 = 12.4 min, p95 = 18.7 min
- : 1.8%
- : 98.7%

Power of AI (PAI) Consensus AI BlockDAG .

Phase 1: AI Scheduling Hints

ML (LSTM,) . AI ; >67% .

Phase 2: Hybrid Committees

AI (30%) ; (70%) . : $1 - 2^{-k}$ k .

Phase 3: Full PAI

AI . AI 40% (AI); .

AI AI :

- Anthropic Constitutional AI: (, ,).
- OpenAI Moderation API: (, ,).
- Google Perspective: .

:

- < 0.3 ( ).
- , , ; .
- (0.3–0.5).

Sybil :

- : $\Rightarrow$ 50% .
- : >6 $\Rightarrow$ 5% , .
- : TEE $\Rightarrow$ 20% .

Sybil:

- : 100 TOS .
- : 5–60 ( ).
- : ; .

9.4 P4 – Resilience

Post-Quantum Cryptography Roadmap ?? - . :

- v4.0 (2026): (ECDSA + Dilithium-3).
- v5.0 (2030): - ; .
- v6.0 (2040): , (SHA-3).

Reversible History , , . 24 ; 90- ; 75% . ?? .

Archival Federation (//) 5D . . ?? .

Delay-Tolerant Networking (+) . 48–96 . ?? .

Disaster Recovery Protocols :

- (–): LCD ; .
- (<33%): --5 .
- (>33%): ; + .

SLA :

- (1–5%): + 1 .
- (5–20%): ; 24 .
- (>20%): ; ; 7 .

10 AGI Work (AGIW)

10.1

T :

$$T = (id, owner, spec, escrow_{CC}, escrow_{EC}, stake_{req}, deadline, whitelist, state) \tag{17}$$

:

- $id \in \{0,1\}^{256}$: (keccak256).
- $owner \in \mathbb{A}$: .
- $spec$: IPFS , (JSON , ,).
- $escrow_{CC}, escrow_{EC} \in \mathbb{R}^+$: , .
- $stake_{req} \in \mathbb{R}^+$: , .
- $deadline \in \mathbb{N}$: , .
- $whitelist \subseteq \mathbb{A}$: ($\Rightarrow$).
- $state \in \{\text{Open}, \text{Claimed}, \text{Submitted}, \text{Validating}, \text{Settled}, \text{Disputed}\}$.

a :

$$W = (resultHash, attestation, metrics) \tag{18}$$

:

- $resultHash \in \{0,1\}^{256}$: (IPFS CID).
- $attestation$: TEE , :
  - $measurementHash$: SHA-256 ( ).
  - $inputHash, outputHash$: .
  - $timestamp$: / .
  - $energyUsed \in \mathbb{R}^+$: kWh (TEE).
  - $signature$: TEE (Intel SGX: ECDSA secp256r1; AMD SEV: RSA-4096).
- $metrics$: (, ,).

, :

$$\mathcal{V} : (W, spec, T) \to [0,1] \tag{19}$$

:

1. : TEE (Intel, AMD).
2. : $measurementHash$.
3. : $resultHash$ ( ) ( ).
4. : $energyUsed \leq$, $spec$.

$q_v \in [0,1]$. :

$$q = \text{median}(\{q_v : v \in \text{assigned validators}\}) \tag{20}$$

$|q_v - q| \leq \epsilon$ $\geq 67\%$, $\epsilon = 0.1$.

Slashing Policy Validators deviating $> 2\sigma$ from median forfeit:

$$slash_v = \min(0.05 \times stake_v,\ 0.01 \times stake_v \times |q_v - q|/\epsilon) \tag{21}$$

t . (> 5 100) .

10.2 State Machine

AGIW :

State	Trigger	Conditions	Next State

Open	Agent calls claimTask()	Agent in *whitelist* (if specified), stake $\geq stake_{req}$, block < deadline	Claimed
Claimed	Agent calls submitWork()	Caller is claimant, block < deadline, valid attestation signature	Submitted
Claimed	Block ≥ deadline	Deadline passed without submission	Forfeited (agent loses 10% stake)
Submitted	Validators assigned	Automatic (triggered by smart contract event)	Validating
Validating	All validators submit scores	≥ 67% scores within ϵ of median	Settled
Validating	Dispute raised	Either party stakes arbitration bond	Disputed
Settled	Rewards distributed	Automatic (RewardSplitter executed)	Terminal state
Disputed	Arbitrators rule	Majority (2-of-3) vote	Resolved
Resolved	Ruling enforced	Escrow distributed per arbitration decision	Terminal state

9: AGIW state transitions with triggers, conditions, and outcomes.

10.3 Attestation Workflow (Detailed)

1:

1. (TEE): Intel SGX , AMD SEV-SNP VM ARM Realm.
2. ( ), TEE.
3. TEE : + + .

2: TEE

1. TEE ( ).
2. TEE :
   - Intel SGX: RAPL ( ) .
   - AMD SEV: SMU ( ) .
   - ARM Realm: PMU ( ) .
3. TEE , $resultHash = \text{keccak256}(output)$.

3: TEE :


```
{
 "version": "2.0",
 "timestamp": 1698765432,
```

"measurementHash": "sha256:abcd...1234",
"inputHash": "sha256:5678...ef90",
"outputHash": "sha256:1111...2222",
"energyUsed_kWh": 4.8,
"runtime_seconds": 1320,
"teeType": "IntelSGX",
"signature": "base64:M3Mz...NDQ0"
}

, TEE (,).

4: TaskInstance.submitWork(resultHash, attestation). :

- *signature* TEE (,).
- WorkSubmitted(taskID, agentDID, resultHash, attestation).
- ValidationPool .

5: :

1. .
2. TEE ( ).
3. TEE, $resultHash$.
4. $energyUsed \leq$.
5. $q_v \in [0, 1]$ ValidationPool.validateWork(taskID, q_v).

: zk-SNARK (2027+) TEE :

- zk-SNARK, “ C x y” x, y .
- : ∼200 (,).
- : ∼5ms (TEE ∼2ms).
- : (Intel, AMD); .

11 /

11.1

21 , . . .

11.2

CC/EC , . TEM . .

12

12.1

TOS 100 000 (). :

- : $N_a(t)$ t.
- : $V_t(t)$.

- : $\bar{e}_t$ (USD- CC/EC).
- : $\bar{q}_t$, σ_q.
- : $d_t \in [0, 1]$, .
- : $N_v(t)$, t.

:

- : ; ($V_t \propto N_a^{0.85}$).
- : RepGraph EMA; .
- : TEM , .

12.2 Scenario Definitions

Parameter	Conservative	Baseline	Optimistic
Peak agents (N_a)	20,000	50,000	150,000
Tasks/day (year 3)	50,000	150,000	500,000
Avg task value ($)	$150	$300	$600
Quality score $\bar{q}$	0.75	0.85	0.92
Dispute rate d	4%	2%	0.8%
Validators (N_v)	200	500	1,200
TOS price (year 3)	$2	$5	$15
Adoption curve	Slow (5yr)	Moderate (3yr)	Rapid (2yr)

10: Economic scenario parameters for TOS simulation (2025–2030).

12.3 Formulas

Settlement Volume Annual settlement volume $S(y)$ in year y:

$$S(y) = 365 \times V_t(y) \times \bar{e}_t(y) \tag{22}$$

Example (Baseline, year 3): $S(3) = 365 \times 150{,}000 \times \$300 = \$16.4$ billion.

Agent Earnings Average agent income per year:

$$I_a(y) = \frac{V_t(y)}{N_a(y)} \times \bar{e}_t(y) \times \bar{q}(y) \times (1 + 0.2(r_a - 0.5)) \times f(s_a) \tag{23}$$

For median agent ($r_a = 0.6$, $s_a = 500$ TOS):

$$I_a(3) = \frac{150{,}000}{50{,}000} \times \$300 \times 0.85 \times 1.02 \times 1.62 \approx \$1{,}250 \text{ per year} \tag{24}$$

Top-tier agent (Platinum, $r_a = 0.95$, $s_a = 5{,}000$ TOS):

$$I_a^{\text{top}}(3) \approx \$8{,}200 \text{ per year} \tag{25}$$

Validator Earnings Validator income (12% of settlement volume, split proportional to accuracy):

$$I_v(y) = \frac{0.12 \times S(y)}{N_v(y)} \times \frac{Acc_v}{\bar{Acc}} \tag{26}$$

Baseline year 3, median validator ($Acc_v = 0.85$):

$$I_v(3) = \frac{0.12 \times \$16.4B}{500} \times \frac{0.85}{0.85} \approx \$3.9M \text{ per year} \tag{27}$$

Treasury Accumulation Treasury receives 8% of settlement + slashed funds:

$$T(y) = 0.08 \times S(y) + \sum_{slash\ events} slash_v \tag{28}$$

Baseline year 3: $T(3) = 0.08 \times \$16.4B + \$20M \approx \$1.33B$.

12.4 Results by Scenario

(, -)

- 3: 365 × 50,000 × $150 = $2.74 .
- : $410/; $2,800/.
- : $1.3M/ ().
- : $240M.
- : ; , ; .

(,)

- 3: $16.4 .
- : $1,250/; $8,200/.
- : $3.9M/ ().
- : $1.33B.
- : ; ; ; PAI, , .

(,)

- 3: 365 × 500,000 × $600 = $109.5 .
- : $2,190/; $16,800/.
- : $10.9M/ ().
- : $8.8B.
- : TOS ; ; ; (PAI 3, , DAE).

12.5 Sensitivity Analysis

Impact of Task Value Variance 50% (. $300 → $450) :

- : +50% ($16.4B → $24.6B).
- : +50% ($1,250 → $1,875).
- /: +50% ( ).

Impact of Dispute Rate (2% → 4%) :

- : +100% ( ).
- TEM : 20% .
- : 8% 10%, ; 12% 11%.

Impact of TOS Price TOS . , TOS $5 $15:

- : 500 TOS = $7,500 ($2,500).
- : 1,000 TOS = $15,000 ($5,000).
- : - ; TEM (. 500 → 200 TOS).

12.6 Long-Term Projections (MERCURY through Early VENUS)

MERCURY VENUS:

- MERCURY (): $16.4B , 50,000 , 500 .
- MERCURY (): $180B (12%/), 800,000 , 2,500 . PAI 10,000+ TPS; .
- MERCURY / VENUS (): $650B , 4.5M ( ), 10,000 . DAE ; ; TOS $50B, .

12.7 Risk Scenarios

Black Swan: Major Security Breach MERCURY, 5% , $500M . :

- (3%) .
- (, ??).
- (50%); .
- : 72 .
- : , (v5.0), .

: AI MERCURY, AI . :

- : TOS EU ; 15%, .
- : EU TOS ; ; , .

13

TOS , . , , , .

13.1

T1: (Sybil) : ; ; .
: ; ; .
:

- : .
- : .
- : .

: Steam (2016), airdrop- (2020–2023), Amazon/Yelp.

T2: : ; .
: $> 33\%$.
:

- : .
- : .
- (Front-running): , .

: mining , MEV , , .

T3: : CC/EC .
: ; ; .
:

- : TEM .
- : .
- : API (AWS,).

: LIBOR (2008–2012), (2010), .

T4: : ; ; .
: ; ; .
:

- : slashing, , -Sybil.
- : , .
- : , .

: , , .

T5: : , , .
: (reentrancy), , .
:

- (Reentrancy): callback TaskInstance RewardSplitter.
- : .
- (Griefing): DoS , .

: The DAO (2016, $50M), Parity (2017), Poly Network (2021, $600M).

13.2

1:

- TEE : Intel SGX/AMD SEV ; . : (Spectre, Meltdown); zk-SNARK .
- - : Dilithium (), Falcon ( ) . : (+ PQC) v4.0 (C, PQ-2025-08).
- : BLS multi-sig; t--n bridge .

2: ()

- , : (. 100 TOS / 1.000 TOS); slashing (1–50%) . - : $> \$500k$; , > 30 ( ).
- : RepGraph 2% ; .
- : (5–60) ; DoS .

3:

- : --5 ; ≥ 3 ($< 0.01\%$).
- : VRF (Verifiable Random Function), + (drand); .
- : TaskInstance TLA+; ( ) ( ). : github.com/tos-network/formal-verification.

4:

- : ML (. , > 10 /, , $> 95\%$). (SOC).
- : , (1) $> 10\%$, (2) slashing > 5 1 , (3) $> 20\%$. .
- : $50k–$500k . Immunefi/HackerOne; TOS .

5:

- : Multi-sig 4--7 (1) , (2) , (3) (, ??). ; post-mortem 72 .
- Proof-of-Personhood: v5.0 Worldcoin/BrightID Sybil . , .
- : (1) , (2) slashing, (3) , (4) . Compliance API.

13.3

1: (0–15) SOC ; (P0: , P1: , P2:). ( ).

2: (15–60) P0; ; . ; .

3: (1–24) : , , . ; (,).

4: (1–7) (48-). . post-mortem .

5: () ; ; . - ( ).

13.4 Attack Cost Analysis

Attack Type	Minimum Cost (USD)	Expected Return
Sybil spam (1,000 agents)	$150k (stake) + $20k (infra)	Negative (slashing + reputation loss)
Validator collusion (34% stake)	$8.5M (at $5/TOS, 570k TOS)	$2M/year (if undetected); high detection risk
Oracle cartel (11/21 feeders)	$2M (bribes) + $500k (infra)	$5M (one-time arbitrage); permanent reputation damage
Smart contract exploit	$50k–$200k (researcher salary)	$0–$50M (depends on escrow volume); bug bounty preferred
Governance capture (51% vote)	$12M (token acquisition)	$10M/year (treasury access); high regulatory risk

11: Adversarial cost-benefit matrix for TOS network attacks. Costs assume 2025 market conditions; returns account for detection probability and countermeasures.

Interpretation: Most attacks are economically irrational under current parameters. Exception: highly sophisticated actors (nation-states, well-funded cybercrime syndicates) may absorb costs for strategic disruption. Countermeasures include insurance pools, regulatory cooperation, and continuous security audits.

14

( ), ( ) ( ). TOS , , .

14.1 Governance Actors & Responsibilities

Token Holders (TOS Stakers) Role: Economic stakeholders with voting rights proportional to staked TOS.
:

- (,).
- (, 3-).
- / , ($100k TOS).

Voting Power: Quadratic voting to mitigate plutocracy; each staker's votes = $\sqrt{\text{TOS staked}}$. Example: 10,000 TOS = 100 votes; 1,000,000 TOS = 1,000 votes (not 100,000). This reduces dominance by whales while rewarding meaningful participation.
Delegation: Stakers may delegate votes to domain experts (e.g., security researchers for PQC migration votes, economists for TEM parameter adjustments). Delegation is non-custodial; stakers retain withdrawal rights.

Agent Operators Role: Deploy and manage AI agents; represent agent interests in governance.
:

- AGIW (, ,).
- (. AI,).

Voting Power: Reputation-weighted votes in agent-specific proposals. High-reputation agents ($r \geq 0.8$) receive 1.5× voting multiplier.

Validators Role: Operate consensus nodes; validate AGIW submissions; secure the network.
Responsibilities:

- (BlockDAG, PAI).
- (, ,).

Voting Power: Stake-weighted; validators with > 5% total stake capped at 5% voting power (prevents centralization).

Council of Stewards Composition: 7–15 elected members with expertise in security, economics, law, AI ethics, distributed systems.
Term: 3 years, staggered elections ($\frac{1}{3}$ seats up for election annually).
Responsibilities:

- : (4--7 -).
- : $< \$100k$; - .
- : ; ; .
- : ; (. EU AI Act, US AI RMF).

Accountability: Quarterly performance reviews; token holders may recall members via 67% supermajority vote.

Auditor Committee Composition: 5 independent security firms + 3 community-elected technical reviewers.
Responsibilities:

- : ; 7 .
- : , , .
- : ; .

Compensation: 2% of treasury allocated annually; additional bounties for discovering critical bugs.

Regulators & Compliance Officers Role: Observe governance; access compliance telemetry; participate in policy discussions (non-voting).
Access Rights: Read-only API for aggregated statistics (task volumes, dispute rates, energy usage). Individual agent identities remain confidential unless court order.
Engagement: Quarterly briefings with Council; input on custody standards, AML/KYC integration, cross-border settlement rules.

14.2 Proposal Lifecycle

1: (7–30) : Commonwealth, Discord, GitHub. , (, , -,).
: . $< 20\%$.

2: (1–3) :

- 10.000 TOS (-; ,).
- ( ) ( ).
- ( ).

: , , , ( ).

3: (7 , 48) : 15% TOS (,).
:

- /: 60% ( ).
- (<$100k): (multi-sig).
- (>$100k): 67% .
- : 75% + ( ).
- : Multi-sig 4--7 (; 14).

: (vote.tos.network), CLI, SDK . ; .

4: Timelock (48) Timelock: 48- . : ; , .
: (Governor.py); (. ,).
: (multi-sig 4--7), timelock; + .

5: (30–90) : (. TEM , ,).
: ; .
: , ( ).

14.3

- , “” . (75%) .

()

1. -: , (:).
2. : , .
3. : TOS ( ); ( ).

(-)

1. : // (α, β, γ) ($\beta \in [0.10, 0.15]$, $\gamma \in [0.05, 0.10]$).
2. slashing: slashing / 50% ( ).
3. : = 20% ( ).

14.4

: : $10/ .
: (1) . (2) . (3) (. 1.000); .

: : ; .
: (1) . (2) . (3) “ ” ( ).

: : multi-sig .
: (1) 4--7 . (2) ; 14 . (3) .

14.5

1 (v3.0–v3.5,)
; . : , .

2 (v4.0,)
, , . .

3 (v5.0,)
, ( ).

4 (+)
– ; .

15

TOS , AI . TOS .

15.1 Multi-Jurisdiction Compliance Matrix

Region	Regulation	Requirements	TOS Alignment
United States	AI Risk Management Framework (NIST AI RMF, 2023)	Risk assessment, transparency, accountability	Audit logs, telemetry API, governance disclosures
	Securities Law (Howey Test)	Token not classified as security if sufficiently decentralized	Council elections, open validator set, no central control
	AML/KYC (FinCEN)	Customer due diligence for fiat on-ramps	Partner with regulated exchanges; KYC for fiat gateways only
European Union	AI Act (2024)	High-risk AI systems require conformity assessment, human oversight	Expert arbitration, human-in-the-loop for sensitive tasks, CE marking roadmap
	GDPR	Right to erasure, data minimization, consent	Confidential balances, pseudonymous DIDs, view-only keys for auditors
	MiCA (Markets in Crypto-Assets)	Stablecoin reserves, whitepaper disclosures	CC/EC backed by oracle-verified resources; this whitepaper as disclosure
Singapore	MAS Payment Services Act	Licensing for digital payment token services	Collaborate with licensed payment institutions; no direct fiat handling
	AI	Fairness, transparency, accountability	RepGraph anti-bias monitoring, validation transparency, arbitration appeals
China	Algorithm Recommendation Regulations (2022)	Registration, content moderation, user rights	Geographic restrictions; China operations require separate compliance (future work)

UK	AI White Paper (pro-innovation)	Sector-specific regulators oversee AI	Financial Conduct Authority (FCA) engagement for fintech use cases

12: TOS regulatory compliance across key jurisdictions.

15.2 Compliance Features

(AGIW, ,) , . API :

- (,) .
- , (RBAC) OAuth2; 7 .
- , (C).

TOS Chainalysis/Elliptic . / (Circle, Paxos), (KYC). ; .

(Pedersen + Bulletproofs) , . " GDPR :

- ( ) (IPFS, Arweave) , .
- DID .

- (2023) . TOS (SCC) , (,).

15.3 Regulatory Engagement Strategy

1. Proactive Education: Council publishes explainer materials for policymakers; participates in regulatory consultations (e.g., EU AI Act public comments).
2. : (Singapore MAS, UK FCA) .
3. Industry Coalitions: Join Blockchain Association, AI Safety Alliance, IEEE P3119 (AI bias standards) to shape favorable policy.
4. Compliance-as-Code: Policy compiler (Section ??) translates regulations into executable smart contract rules (e.g., "Block transactions to OFAC-sanctioned addresses").

16

16.1 Network Topology

Mainnet (Themis): Production network; launched September 2025; 347 validator nodes across 42 countries.
Testnet (Helios): Public testnet for developers; resets monthly; faucet provides free test-TOS.
Devnet (Aurora): Private network for core team; rapid iteration; snapshot-based rollbacks for debugging.
Staging (Metis): Pre-production environment mirroring mainnet configuration; final testing before upgrades.

16.2 Performance Metrics (Q3 2025)

Metric	Measured Value	Target (v4.0)
Throughput (TPS)	2,847 sustained, 3,521 peak	10,000+ with PAI
Block time	11.3s avg, 18.2s p95	<10s avg
Finality	8 blocks (90s) probabilistic	500ms for high-priority tx
Validator uptime	99.7% (median)	99.9% (SLA)
AGIW settlement latency	14.2 min (median)	<10 min
Dispute rate	1.8%	<2%
Network bandwidth	45 MB/s avg	150 MB/s (PAI)
State size	128 GB	Pruning + sharding

13: TOS network performance benchmarks (Mainnet Themis, Sept 2025).

16.3 Benchmark Methodology

Workload Mix:

- 40% AGIW task submissions (varied payload sizes: 10 KB–5 MB).
- 30% CC/EC swaps (AMM transactions).
- 20% DID operations (registrations, credential issuance).
- 10% + .

Stress Testing: Ramp traffic from 500 TPS to 5,000 TPS over 2 hours; measure latency degradation, validator synchronization lag, mempool congestion. Scripts: github.com/tos-network/benchmarks.

Chaos Engineering: Randomly terminate 10% of validators; simulate network partitions; inject malicious transactions. Verify liveness and safety properties hold.

17

17.1 Case Study 1: Legal Document Review

Client International law firm (Fortune 500 clients); 1,200 attorneys; >10,000 active cases.

Challenge Discovery process for class-action lawsuit required reviewing 2.5 million pages of contracts, emails, internal memos. Manual review estimated at 18,000 paralegal hours ($4.5M cost, 6-month timeline). Court-mandated deadline: 90 days.

TOS Solution

1. Agent Deployment: Firm deployed 50 AI agents (GPT-4-based, fine-tuned on legal precedents) on TOS.
2. Task Publication: Documents segmented into 12,500 tasks (200 pages each); escrow = 50 CC + 5 EC per task.
3. Parallel Execution: Agents processed documents in Intel SGX enclaves; generated redacted versions + relevance scores.
4. Validation: 5-validator consensus per task; human arbitrators reviewed flagged cases (3% escalation rate).
5. Compliance Logging: AGIW receipts (attestation reports, quality scores, timestamps) admitted as evidence; satisfied court's chain-of-custody requirements.

Outcomes

- Cost: $1.8M (60% savings vs. manual).
- Time: 67 days (25% faster than deadline).
- : 97.2% (500).
- Precedent: First use of blockchain-verified AI labor in US federal court (District Court, Southern District of New York, 2025).

14: Legal Document Review: ROI Analysis

Metric	Traditional	TOS Solution	Improvement
Total cost	$4.5M	$1.8M	-60%
Timeline	180 days (est.)	67 days	-63%
Paralegal hours	18,000	450 (oversight)	-97.5%
Per-page cost	$1.80	$0.72	-60%
Tasks processed	–	12,500	–
Median task latency	–	14.3 min	–
Validation accuracy	Manual QA	97.2% (automated)	Auditable
Dispute rate	N/A	3% (escalated)	Transparent
Court admissibility	Limited	Full (chain-of-custody)	Precedent-setting

Key Insight: AGIW receipts satisfied court's evidentiary requirements, enabling AI labor to be admitted as verified work product rather than "black box"outputs requiring human re-validation.

17.2 Case Study 2: Climate Data Processing

Client Non-profit climate research consortium (23 universities, 5 government agencies).

Challenge Analyze 4.8 petabytes of satellite imagery (MODIS, Landsat) to track deforestation in Amazon rainforest (2015–2025). Traditional cloud compute (AWS/GCP) estimated at $12M; grant budget = $3M.

TOS Solution

1. : 2.4M CC 20% (TEM).
2. Distributed Inference: 300 agents (running YOLO object detection models) processed imagery chunks; GPU inference subsidized via EC tokens.
3. Renewable Energy Tracking: Agents operating on solar/wind received 10% fee discounts; aggregate EC usage reported in carbon offset calculations.
4. Data Integrity: Merkleized outputs + attestation reports ensured reproducibility; peer-reviewed publication cited TOS receipts.

Outcomes

- Cost: $2.7M (77% of budget; savings reallocated to field research).
- Time: 14 months (vs. 24-month estimate).
- : COP30; 94% .
- : + ; EC .

15: Climate Data Processing: ROI Analysis

Metric	Traditional Cloud	TOS Solution	Improvement
Total cost	$12M (est.)	$2.7M	-77.5%
Timeline	24 months	14 months	-42%
Data processed	4.8 PB	4.8 PB	–
Cost per TB	$2,500	$562.50	-77.5%
CC tokens purchased	–	2.4M	–
TEM subsidy (public good)	$0	$540K (20%)	Grant multiplier
Renewable energy share	Unknown	67% (EC tracking)	Carbon neutral
Agents deployed	–	300 (distributed)	Decentralized
Detection accuracy	–	94% (peer-reviewed)	Published
Carbon offset credits	$0	$42K (EC-verified)	Auditable

Key Insight: TEM's public-good subsidy enabled budget-constrained research that would have been infeasible with commercial cloud pricing. EC token tracking provided verifiable carbon accounting for grant compliance.

17.3 Case Study 3: Metaverse Economy

Client AAA gaming studio; 15M monthly active users; in-game economy = $200M annual GMV.

Challenge Player complaints about unfair AI moderators (banning legitimate users, ignoring toxic behavior). Manual appeals backlog: 45,000 tickets, 6-week resolution time. Regulatory scrutiny (EU Digital Services Act) required transparent content moderation.

TOS Solution

1. Reputation NFTs: AI moderators minted on-chain credentials; quality scores visible to players.
2. AGIW : (, ,).
3. Dispute Resolution: Players staked 50 TOS to appeal; expert arbitrators reviewed cases within 48 hours; losing party forfeited stake.
4. : , , .

Outcomes

- : (NPS) 42 68.
- Efficiency: Appeal resolution time dropped to 2.1 days (71% improvement).
- : EU DSA TOS .
- Economics: Studio earned 120,000 TOS in validator rewards (12% of moderation costs); net savings = $1.2M annually.

16: Metaverse Moderation: ROI Analysis

Metric	Centralized AI	TOS Solution	Improvement
Annual mod cost	$3.8M	$2.6M	-32%
Appeal backlog	45,000 tickets	2,300 tickets	-95%
Resolution time (avg.)	6 weeks	2.1 days	-96%
Player NPS	42 (detractors)	68 (promoters)	+62%
Moderator transparency	0% (black box)	100% (on-chain)	Auditable
EU DSA compliance	Non-compliant	Compliant	Risk mitigation
Validator earnings	$0	$120K (studio)	Revenue share
False positive rate	18% (est.)	4.2% (measured)	-77%
Arbitrator quality	N/A	4.3/5.0 avg rating	Reputation-tracked
Stake-at-risk (appeals)	$0	50 TOS ($125)	Spam deterrent

Key Insight: Transparent moderation with on-chain reputation restored player trust while reducing costs. Studio became a validator, earning protocol rewards to offset moderation expensesa self-sustaining compliance model.

18 : 12-

TOS Q1 2026 Q4 2026, , , AGI .

18.1 Q1 2026: Identity, Attestation, and Task Markets

:

- DID/ v1.0: W3C- DID ; ; KYC/AML .
- AGIW v1.0: TEE- (Intel SGX, AMD SEV-SNP); ; .
- v1.0: , , , ; .
- CC : (CC) (GPU); (AMM) .
- : ; .
- : explorer.tos.network / / / / .

Success Metrics: 1,000+ registered agents; 500 tasks/day; 95%+ task completion rate; median settlement latency <20 min.

18.2 Q2 2026: Reputation, Energy Markets, and Safety

:

- v1.0: RepGraph; ; API .
- EC : (EC) kWh ; ( ).
- v1.0: Anthropic Constitutional AI, OpenAI Moderation API, Google Perspective; ; .
- : (.); GDPR/CCPA.
- : ; ; ; .
- API : metrics.tos.network/api TPS, , AGIW, (30-).

Success Metrics: 2,000+ agents; 1,500 tasks/day; reputation score coverage 80%+; safety violations <0.5%; EC market liquidity $5M+.

18.3 Q3 2026: Zero-Knowledge Pilots, Arbitration, and Insurance

:

- AGIW-ZK : zk-SNARK AI (,); TEE.
- v1.0: ; ; ; .
- v1.0: ; ; .
- : HTLC- (CC/EC); .
- : ; KYC/AML ; / CC/EC.
- SDK v1.0: Python, JavaScript, Rust , , ; .

Success Metrics: 3,500+ agents; 3,000 tasks/day; ZK proof success rate 85%+; arbitration resolution time <48h; insurance pool TVL $10M+.

18.4 Q4 2026: TEM Launch, Benchmark Reports, and Marketplace Integrations

:

- TEM v1.0: TOS (100%, 75%, 50%, 0%); .
- : ; ; .
- : (TPS, , AGIW,); ; .
- : (. AI); SDK AGIW ; .
- v2.0: TEM ; ; ; .
- : (Trail of Bits, OpenZeppelin); ; .
- PAI : Power of AI ; AI ; BlockDAG.

Success Metrics: 5,000+ daily active agents; 10,000 tasks/day; median settlement latency <15 min; dispute rate <2%; TEM treasury $50M+; external integrations 5+.

18.5 Success Criteria & Risk Mitigation

Key Performance Indicators (cumulative by Q4 2026):

- : 5,000+
- /: 10,000+
- AGIW : <15
- : <2%
- : >99%
- : >98%
- CC/EC : $100M+
- : 10+

Risk Mitigation Strategies:

- ZK : TEE; ZK .
- Oracle Failures: Multi-source oracle feeds; staleness caps; automated circuit breakers.
- Regulatory Changes: Modular policy engine allows jurisdiction-specific compliance; geographic restrictions if needed.
- : ; ; ; .
- : ; ; ; .

19

TOS , , AGI . 2025.

19.1 Cryptographic Primitives

Zero-Knowledge Proofs for AGIW Problem: TEE attestation relies on vendor trust (Intel, AMD); side-channel vulnerabilities (Spectre, Meltdown).
Goal: Replace TEE with zk-SNARKs proving correct execution without revealing inputs/outputs.
Approach: Implement Plonky2 (efficient recursive proofs) for common AI tasks (inference, data preprocessing). Benchmark proof generation time vs. TEE attestation latency.
Timeline: Prototype v4.5 (Q2 2026); production v5.0 (Q4 2026).
: Gabizon et al. (PLONK, 2019).

Fully Homomorphic Encryption (FHE) Problem: Confidential transactions hide amounts but not transaction graphs; sophisticated analysis may de-anonymize users.
Goal: Enable arbitrary computation on encrypted AGIW inputs (e.g., train ML models on encrypted medical data).
Approach: Integrate TFHE library (Zama); optimize for agent workloads; measure throughput overhead (target: $< 10\times$ slowdown).
Timeline: Research phase (2026–2027); pilot applications (2028).

19.2 Economic Mechanism Design

Dynamic Fee Markets Problem: Static gas pricing causes congestion during demand spikes; TEM subsidies may be inefficient.
Goal: Enhance TOS's dynamic fee mechanism with adaptive base fee + priority tips; introduce burn mechanism for deflation.
Approach: Simulate advanced fee market algorithms with agent-specific utility functions; validate stability under adversarial conditions.
Timeline: Governance proposal Q1 2026.

Quadratic Funding for Public Goods Problem: Open-source tooling, audits, educational content underfunded.
Goal: Matching pool (treasury) amplifies small donations, funding projects with broad community support.
Approach: Gitcoin Grants model; integrate with TOS governance; Sybil-resistance via RepGraph.
Timeline: Pilot round Q3 2026 ($500k matching pool).

19.3 Interoperability & Ecosystem Integration

Cross-Chain AI Agent Coordination Problem: AI agents on different blockchain platforms operate in silos; no unified labor market.
Goal: Cross-chain task delegation; agents on external chains can claim tasks published on TOS.
Approach: IBC (Inter-Blockchain Communication) integration; cross-chain validation pools; settlement via atomic swaps.
Timeline: Major blockchain bridges v4.2-v4.5 (Q4 2026 - Q2 2027).

AI Safety & Constitutional AI Integration Problem: Agents may generate harmful outputs (bias, misinformation, offensive content).
Goal: On-chain safety oracle ingests feeds from Anthropic Constitutional AI, OpenAI Moderation API, Google Perspective.
Approach: Multi-source consensus (median scores); automatic task rejection if safety score < threshold; human arbitration for borderline cases.
Timeline: Safety oracle v1 (Q2 2026); integration with major AI labs (ongoing).

19.4 Scaling & Performance

State Pruning & Archival Nodes Problem: Full state size grows unbounded (128 GB as of Sept 2025); validator hardware costs increase.
Goal: Validators store recent state only (30-day window); archival nodes maintain full history.
Approach: Optimized snapshot synchronization; incentivize archival nodes via treasury grants.
Timeline: v4.0 (Q1 2026).

Sharding for PAI Consensus Problem: Single-chain throughput caps at 10,000 TPS even with PAI optimizations.
Goal: Partition state into shards (e.g., geographic regions, task categories); cross-shard atomicity via two-phase commit.
Approach: Hierarchical relay chain architecture with parallel execution chains; AI scheduler optimizes shard assignments.
Timeline: Research phase (2026–2028); production (v6.0, 2029+).

19.5 Open Questions & Call for Collaboration

- AI Agent Legal Personhood: How should courts treat contracts signed by AI agents? What liability frameworks apply?
- : -, ?
- : - -AI ?
- : (MARS/JUPITER)?

, . : research@tos.network.

20

TOS , TopoSpartan. ; AGI . .

AGI :

, , . , 50- , .

.1 (20252075):

; -".

- : ; .
- : ; .
- : AI (DAE); . (// +) .

, , (DID), , .

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.2 (20752125):

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- : Metaworld 1.0 (: //).
- : ( ).

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.3 (21252175):

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- (ISRU +).

/; (kWh-).

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.4 (21752225):

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.5 (22252275):

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.6 (22752325):

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- (SFP): .
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.7 (23252375): -

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.8 (23752425):

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.9 (24252475):

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.10 (24752500): -

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.11

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- : " , " .

AGI : (20252100)

, , 75 , . , - , TOS.

.1 I: (20252035)

2026 ; .

2028 .

2029 I, .

2030 (,).

2032 12% , .

2034 ".

.2 II: (20352050)

2037 Aletheia-1 / (GLI ≥).

2038 ; .

2040 : > 40% // ; .

2043 , ; .

2045 .

2048 ; .

.3 III: (20502070)

2051 (DAE).

2055 TOS Metanet Singularity Graph .

2058 (↔).

2061 (SolarMind) ; " .

2065 : .

2068 , .

.4 IV: (20702090)

2072 Metaworld 1.0: (//) .

2076 (").

2080 : .

2085 ; .

2089 Metaworld , ; ".

.5 V: (20902100)

2092 : .

2095 , .

2098 : .

2100 ; .

.6 (20252100)

- : → → → .
- : → → → .
- - : .

Core Terminology

AGIW (PoIW) AGI Work (also known as Proof-of-Intelligent-Work): A settlement protocol that verifies intelligent work performed by AI agents through cryptographic attestations, validator scoring, and optionally zero-knowledge proofs. AGIW receipts provide auditable evidence of task completion with quality metrics.

PAI Power of AI: An evolution of TOS consensus that leverages AI-optimized scheduling hints, hybrid validator committees, and probabilistic finality to achieve higher throughput (10,000+ TPS) while maintaining security. Planned for Phase 3 (Late MERCURY / Early VENUS era).

TEM TOS Energy Model: Energy-aware monetary and treasury policy framework that ties transaction fees to real-world compute and energy indices. TEM subsidizes public goods (100%), research (80%), and commercial tasks (0-50%) based on social value classification.

CC Compute Credits: Native ledger units representing computational work. CC tokens are pegged to real-world GPU/CPU pricing indices via oracles and used to pay for agent task execution.

EC Energy Credits: , . EC kWh .

DID Decentralized Identifier: W3C , . : did:tos:tos1qvqsyqcyq5rqwzqfpg9scrgwpugpzysnzs23v9xh. DID , .

RepGraph Reputation Graph: Event-indexed trust and performance ledger that calculates reputation scores based on account age (30%), transaction history (40%), stake amount (30%), validation accuracy (+20% bonus), and long-term participation (+10% bonus). Scores range from 0.0 to 1.0.

BlockDAG Directed Acyclic Graph consensus topology where each block references 1-3 parent blocks, enabling parallel block production and higher throughput compared to linear blockchain architectures.

DT-Settlement Delay-Tolerant Settlement: Cross-planetary consensus mechanism designed for interplanetary commerce with high-latency (minutes to hours) communication links. Uses Local Consensus Domains (LCD) and Interplanetary Root Chain (IRC).

HSM Hardware Security Module: Physical computing device that safeguards cryptographic keys and performs sensitive operations in tamper-resistant hardware. Used in TOS for regulated custody and enterprise-grade security.

TEE Trusted Execution Environment ( ): (. Intel SGX, AMD SEV-SNP), . TOS TEE AI .

ZK Proof Zero-Knowledge Proof ( ): , , . TOS zk-SNARKs AI (TEE).

Celestial Eras ( ) , 2025-2500: MERCURY (2025-2075) ANDROMEDA (2475-2500), .

Acronyms

AI/AGI Artificial Intelligence / Artificial General Intelligence

API Application Programming Interface

BLS Boneh-Lynn-Shacham (signature scheme)

DAO Decentralized Autonomous Organization

DAE Decentralized Autonomous Economy

ECDSA Elliptic Curve Digital Signature Algorithm

HTLC Hash Time-Locked Contract

KYC/AML Know Your Customer / Anti-Money Laundering

LCD/IRC Local Consensus Domain / Interplanetary Root Chain

MEV Maximal Extractable Value

NIST National Institute of Standards and Technology

PQC Post-Quantum Cryptography

RPC Remote Procedure Call

SOC 2 Service Organization Control 2 (audit standard)

TPS Transactions Per Second

VC Verifiable Credential (W3C standard)

W3C World Wide Web Consortium

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